Noel Alex Samuel

AI / ML ENGINEERING · DATA SCIENCE · DATA SYSTEMS

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Work authorization: India (OCI), Saudi Arabia (Premium Residency) Available Jan 2027 for a full-semester onsite internship

PROFILE

Final-year CSE (Data Science) student with six months of paid AI engineering in retrieval, evaluation and data systems.

PROFESSIONAL EXPERIENCE

AI Engineer (contract) — Blackridge Technologies / Stravah Mar–Sep 2026

- Built AI workflows that extract and reconcile transactions from complex Indian bank-statement PDFs: layout and table reconstruction, OCR, row recovery and ledger consistency checks.
- Served GLM-OCR on GPUs with FastAPI, vLLM and Modal (L40S), tuning batching, concurrency, caching, observability and cost, backed by load and quality tests.

SELECTED PROJECTS & RESEARCH

Arachne + PhishGCN — Distributed web capture and phishing detection 2025–26

- Built a Rust crawler and capture platform on NATS JetStream, SQLite and content-addressed zstd storage, with headless Chrome, per-hop SSRF checks and volunteer workers over Tailscale; 203,367 captures in the Oct 2026 snapshot.
- Showed most phishing-feed URLs no longer serve a phishing page when fetched (64.7% rule-flagged as stand-ins); training on hand-audited labels lifted PR-AUC from 0.614 to 0.724 (95% CI 0.668–0.780) on 1,196 audited captures, with 61.4% recall at 0.5% false alarms (site/kit-disjoint folds).

Stretchy + HALO — Evidence-grounded retrieval platform and embedding research 2025–26

- Leading a fault-tolerant Python/Rust pipeline: extraction, PostgreSQL/Temporal orchestration, Qdrant retrieval, provenance.
- Inventoried 862,917 files (130.26 GB) in one run with zero read errors; chose the dense embedder by benchmark (288 units/s at 2.25 GiB peak VRAM); HALO re-encoded 5.1× fewer tokens than late chunking on 2,536 real document edits.

HELIX Lite — Parallelizing compiler in Rust (public, runs in the browser) 2026

- Wrote the whole pipeline (Pratt parser, type checker, SSA, dependence tests, Cranelift x86-64) to prove loops safe before parallelizing: 5.79× saxpy speedup on 8 threads; unsafe loops are refused, naming the dependence.

Sivana — Audio recognition in Rust and WASM (public) 2025–26

- Built a fingerprint engine with a memory-mapped index and in-browser WASM fingerprinting; a calibrated gate recognises 76.7% of degraded clips at zero false accepts (prototype: 35%), at 0.4 ms per match.

PolySafe / Vergil — Reliable drug-interaction ML 2026

- Led GNN and Chemprop experiments on 63,472 drug pairs and 964 side-effect classes; built drug-disjoint and scaffold splits, leakage gates and calibration checks, and reported where abstention guarantees failed.

Living Review / VCAP — Code-review comments that survive resubmission 2026

- Co-authoring a manuscript: across 7,200 controlled edits and 15,064 real-history observations, missed re-checks fell from 12 to 1 of 2,780 changed targets versus CodeMapper.

PPLC — Privacy layer for LLM context 2026

- Co-developed pseudonymization that keeps personal data out of LLM calls and restores it safely in responses: 0 leaks versus 11,052 for global replacement on an authored 11,792-attack benchmark, with all 4,422 benign responses restored.

EDUCATION

Vellore Institute of Technology, Vellore B.Tech, Computer Science & Engineering (Data Science) · 2023–27

New Middle East International School, Riyadh CBSE (Classes X and XII) · Physics subject topper · 2023

LEADERSHIP & ACHIEVEMENTS

GDG on Campus VIT · Senior Core Member, Technical (AI/ML) (2023–26) Ran AI/ML recruitment: my domain quiz cut 256 shortlisted applicants to about 80, each set a project and interviewed by me; helped run DevJams and Women Techies.

Inside Google's DeepMind (Nov 2024) Gave a recorded session for 400+ students, from transformers to AlphaFold.

Head Boy (2022–23) Led student welfare and represented the school at a regional CBSE principals' conference.

Invention disclosures (2026) Five collaborative disclosures submitted through VIT's IPR process.

TECHNICAL SKILLS & CERTIFICATION

Languages Python, Rust, SQL, TypeScript, C/C++, Java, R

ML PyTorch, Transformers, GNNs, NLP / OCR, RAG, evaluation design, pandas, scikit-learn

Systems FastAPI, vLLM, Modal, PostgreSQL, NATS JetStream, Qdrant, Docker, Temporal, Linux

Certification Oracle Cloud Infrastructure 2025 Generative AI Professional (92%)